

StockMaster - Software Requirements Specification (SRS)

- **Version:** 4.0
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 - **Project:** StockMaster - Professional Inventory Management System
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1. Introduction

1.1 Purpose

This document describes the functional and non-functional requirements for StockMaster v4.0, a desktop inventory management application designed for small to medium businesses.

1.2 Scope

StockMaster provides stock control, point of sale (POS), purchase order management, reporting, user authentication, and bilingual support (Spanish/English) - all running locally without internet dependency.

1.3 Definitions

- **POS:** Point of Sale module for processing customer sales
- **IVA:** Impuesto al Valor Agregado Argentine value-added tax
- **Batch:** A specific lot of a product with its own expiry date and quantity
- **KPI:** Key Performance Indicator - metrics shown on the dashboard
- **i18n:** Internationalization multi-language support
- **ESC/POS:** Epson Standard Code for Point of Sale - thermal printer protocol

2. Overall Description

2.1 Product Perspective

StockMaster is a standalone desktop application built with JavaFX. It uses an embedded SQLite database for local data persistence, requiring no external server or internet connection.

2.2 User Classes

- **ADMIN:** Full access: manage products, users, suppliers, categories, sales, purchase orders, reports, backup/restore, language settings
- **CASHIER:** Limited access: sales (POS), product search, view dashboard

2.3 Operating Environment

- **OS:** Windows 10/11 (64-bit)
- **Runtime:** Java 17+ (bundled via jpackage installer)
- **Database:** SQLite (embedded, file-based)
- **Hardware:** Optional ESC/POS thermal printer for receipts

2.4 Design Constraints

- Must work fully offline
- All data stored locally (privacy by design)
- Single-user at a time (desktop app)
- Installer under 100 MB

3. Functional Requirements

3.1 Authentication (FR-AUTH)

- **FR-AUTH-01:** System shall display a login screen on startup (Priority: High)
- **FR-AUTH-02:** Users authenticate with username and password (SHA-256 hashed) (Priority: High)
- **FR-AUTH-03:** Default admin account created on first run (admin/admin) (Priority: High)
- **FR-AUTH-04:** System tracks current session (username, role, user ID) (Priority: High)
- **FR-AUTH-05:** Logout returns to login screen (Ctrl+Q shortcut) (Priority: Medium)

3.2 Dashboard (FR-DASH)

- **FR-DASH-01:** Display KPI cards: total products, inventory value, low stock count, expiring batches (30 days) (Priority: High)
- **FR-DASH-02:** Display bar chart of product distribution by category (Priority: Medium)
- **FR-DASH-03:** KPI values update on view load (Priority: High)

3.3 Product Management (FR-PROD)

- **FR-PROD-01:** CRUD operations for products (name, description, category, supplier, prices, stock, barcode, tax rate) (Priority: High)
- **FR-PROD-02:** Search/filter products by name, category, or barcode (Priority: High)
- **FR-PROD-03:** Display products in sortable table with columns: ID, Name, Category, Supplier, Buy Price, Sell Price, Stock, Min Stock (Priority: High)
- **FR-PROD-04:** Visual low-stock indicators (row highlighting) (Priority: Medium)
- **FR-PROD-05:** Barcode detection via keyboard input (Priority: Medium)
- **FR-PROD-06:** Import products from CSV file (Priority: Medium)
- **FR-PROD-07:** Export products to CSV file (Priority: Medium)
- **FR-PROD-08:** Export product list to PDF (Priority: Medium)
- **FR-PROD-09:** Print product ticket to ESC/POS thermal printer (Priority: Low)
- **FR-PROD-10:** Open cash drawer via ESC/POS command (Priority: Low)

3.4 Batch/Lot Tracking (FR-BATCH)

- **FR-BATCH-01:** Each product can have multiple batches with: lot number, expiry date, quantity, cost (Priority: High)
- **FR-BATCH-02:** Batch panel appears when selecting a product in the table (Priority: High)
- **FR-BATCH-03:** Add, delete batches inline (Priority: High)
- **FR-BATCH-04:** Display batch status (ACTIVE, EXPIRED) based on expiry date (Priority: Medium)

3.5 Sales / Point of Sale (FR-SALE)

- **FR-SALE-01:** Search products by name or barcode for adding to cart (Priority: High)
- **FR-SALE-02:** Shopping cart with product name, quantity, unit price, tax rate, line total (Priority: High)
- **FR-SALE-03:** Adjustable quantity via spinner control (Priority: High)
- **FR-SALE-04:** Remove individual items or clear entire cart (Priority: High)
- **FR-SALE-05:** Payment summary panel: subtotal, tax (IVA), total (Priority: High)
- **FR-SALE-06:** Automatic tax calculation per item (based on product tax rate) (Priority: High)
- **FR-SALE-07:** Payment method selection: Cash, Credit Card, Debit Card, Transfer (Priority: High)
- **FR-SALE-08:** Complete sale: insert sale record, deduct stock, log stock movements (Priority: High)
- **FR-SALE-09:** Reject sale if insufficient stock for any item (Priority: High)
- **FR-SALE-10:** Recent sales history table (collapsible) (Priority: Medium)
- **FR-SALE-11:** Print receipt to ESC/POS thermal printer (Priority: Low)

3.6 Purchase Orders (FR-PO)

- **FR-PO-01:** Create purchase orders with: supplier, notes, and line items (Priority: High)
- **FR-PO-02:** Each line item has: product, quantity, unit cost, line total (Priority: High)
- **FR-PO-03:** Display orders table: ID, date, supplier, total, status, notes (Priority: High)
- **FR-PO-04:** Order statuses: PENDING, RECEIVED, CANCELLED (Priority: High)
- **FR-PO-05:** Receive order: update status to RECEIVED and add quantities to product stock (Priority: High)
- **FR-PO-06:** Cancel order: update status to CANCELLED (no stock change) (Priority: Medium)
- **FR-PO-07:** View order items in detail (Priority: Medium)

3.7 Reports (FR-RPT)

- **FR-RPT-01:** Report types: Sales Summary, Top Products, Stock Movements, Inventory Value, Profit Margins, Low Stock, Expiring Soon (Priority: High)
- **FR-RPT-02:** Date range filter (from/to) (Priority: High)
- **FR-RPT-03:** Summary cards: total records, total value, average (Priority: High)
- **FR-RPT-04:** Dynamic data table based on report type (Priority: High)
- **FR-RPT-05:** Export report to PDF (Priority: Medium)

- **FR-RPT-06:** Export report to Excel (.xlsx) (Priority: Medium)

3.8 Category Management (FR-CAT)

- **FR-CAT-01:** CRUD operations for categories (name, description) (Priority: High)
- **FR-CAT-02:** Display categories in table with edit form (Priority: High)

3.9 Supplier Management (FR-SUP)

- **FR-SUP-01:** CRUD operations for suppliers (name, contact, phone, email) (Priority: High)
- **FR-SUP-02:** Display suppliers in table with edit form (Priority: High)

3.10 User Management (FR-USER)

- **FR-USER-01:** CRUD operations for users (username, display name, role, password, active status) (Priority: High)
- **FR-USER-02:** Search users by name (Priority: Medium)
- **FR-USER-03:** Prevent deleting own account (Priority: High)
- **FR-USER-04:** Password hashing with SHA-256 (Priority: High)

3.11 Internationalization (FR-I18N)

- **FR-I18N-01:** Full Spanish and English interface (Priority: High)
- **FR-I18N-02:** Language toggle in sidebar (Priority: High)
- **FR-I18N-03:** All view labels, buttons, columns, and messages translated via ResourceBundle (Priority: High)
- **FR-I18N-04:** Language preference persisted (Java Preferences API) (Priority: Medium)

3.12 Backup & Restore (FR-BKP)

- **FR-BKP-01:** Export SQLite database file as backup (Priority: Medium)
- **FR-BKP-02:** Restore database from backup file (replaces current data) (Priority: Medium)
- **FR-BKP-03:** Confirmation dialog before restore (Priority: High)

4. Non-Functional Requirements

4.1 Performance

- **NFR-01:** Application startup in under 3 seconds
- **NFR-02:** View transitions in under 500ms
- **NFR-03:** Search results returned in under 200ms for up to 10,000 products

4.2 Security

- **NFR-04:** Passwords stored as SHA-256 hashes (never plain text)
- **NFR-05:** No external network connections (100% offline)
- **NFR-06:** Database file stored locally with OS-level file permissions

4.3 Usability

- **NFR-07:** Dark-themed modern UI consistent across all views
- **NFR-08:** Sidebar navigation with icons and text labels
- **NFR-09:** Responsive layout adapting to window resize
- **NFR-10:** Bilingual interface (ES/EN) with instant switching

4.4 Reliability

- **NFR-11:** Database transactions with rollback on failure (sales, purchase orders)
- **NFR-12:** Connection pooling for database access
- **NFR-13:** Graceful error handling with user-visible messages

4.5 Portability

- **NFR-14:** Windows 10/11 support via native installer (.exe)
- **NFR-15:** No external dependencies required at runtime (JRE bundled)

5. Technology Stack

- **Language:** Java 17+ (JDK 24)
- **UI Framework:** JavaFX 17.0.2
- **Database:** SQLite 3.x (via sqlite-jdbc)
- **Build Tool:** Maven 3.x
- **PDF Generation:** iTextPDF 5.5.13.4
- **Excel Export:** Apache POI 5.2.5
- **Installer:** jpackage + WIX JDK 14+
- **Thermal Printing:** javax.print (ESC/POS) Java built-in

6. Database Schema

Tables

- **products:** id, name, description, category_id, supplier_id, price_buy, price_sell, stock_current, stock_min, barcode, tax_rate
- **categories:** id, name, description
- **suppliers:** id, name, contact_person, phone, email
- **users:** id, username, password_hash, display_name, role, active
- **sales:** id, sale_date, user_id, subtotal, tax_amount, total, payment_method, status
- **sale_items:** id, sale_id, product_id, product_name, quantity, unit_price, tax_rate, line_total
- **purchase_orders:** id, order_date, supplier_id, supplier_name, total, status, notes
- **purchase_order_items:** id, order_id, product_id, product_name, quantity, unit_cost, line_total
- **batches:** id, product_id, lot_number, expiry_date, quantity, cost_per_unit, received_date, status
- **stock_movements:** id, product_id, type (IN/OUT), quantity, reference, user_id, created_at

- **price_history:** id, product_id, old_price, new_price, changed_by, changed_at
- **audit_log:** id, action, entity, entity_id, user_id, details, created_at

7. User Interface

7.1 Navigation

Sidebar with icon buttons for: Dashboard, Sales, Products, Purchase Orders, Categories, Suppliers, Reports, Users. Bottom section: language toggle, backup/restore, logout.

7.2 Views

1. **Login:** Username/password form with StockMaster branding
2. **Dashboard:** 4 KPI cards + category bar chart
3. **Sales (POS):** product search, cart, payment summary, recent sales
4. **Products:** table + edit form + batch panel
5. **Purchase Orders:** orders table + inline order form with items
6. **Categories:** table + edit form
7. **Suppliers:** table + edit form
8. **Reports / Users:** type selector, date filters, summary cards, data table, export buttons, table + edit form